

AdvocateAuroraHealth

Title: HIT Hardware Asset Management Standard		Document Number: 63322
Document Type: ☐ Policy ☐ Procedure ☐ Guideline ☒ Other Standard		Last Review/Revision Date: 06/01/2021
Content Applies to Patient Care: (Select all that apply) ☐ Adults ☐ Pediatrics (Under 18)	Content Applies to: (Select One) ☐ Clinical ☒ Administrative	Next Review Date: 06/01/2024
		Effective Date: 06/01/2021
Scope: ☐ AAH System ☐ ACH ☐ AAH IL Only ☐ AAH WI Only ☐ Site Only (Location Name): ☒ Department Only (Department Name): HIT		

I. PURPOSE

To establish HIT Hardware Asset Management (HAM) standard requirements for all Advocate Aurora Health team members for the HAM program at AAH.

II. SCOPE

This standard applies to Advocate Aurora Health, Inc. and any entity or facility owned and controlled by AAH. It includes hardware issued to team members and non-team members. The key activities related to all hardware assets are:

- Procurement/ Acquisition
- Nonstandard Procurement/Acquisition
- Deployment and Redeployment
- Management/ Operation
- Reclamation
- Disposal
- Asset Reconciliation

III. DEFINITIONS/ABBREVIATIONS

Capital Asset – Long-lived asset with an estimated useful life of one year or more and a historical cost of at least \$5,000. For purposes of this standard, the term fixed asset has the same definition as capital asset.

Examples include:

- a. Land and land improvements
- b. Buildings and leasehold improvements
- c. Equipment (including HIT hardware)
- d. Computer software
- e. Vehicles

HIT Asset – Any item of value that HIT must track and manage

Asset Tag: - A unique number and barcode assigned and attached to each non-consumable asset by AAH.

Asset Status in Stock - Equipment that is in inventory and available for deployment.

Asset Status in Transit - Equipment that is being shipped and pending install.

Asset Status in Use - Equipment that is active in the production environment.

Business Associate - Shall have the same meaning as the term “business associate” at 45 CFR §160.103.

COD - Certificate of destruction

Change (CHG) - The process responsible for controlling the lifecycle of all changes, enabling beneficial changes to be made with minimal disruption to HIT services and team members.

Chargeback – HIT will purchase the item from their budget and then perform a general ledger transfer to the final budget area based on the approval obtained from the leader in the Service Now request.

Configuration Item (CI) - Any computer, device, software, or service in the CMDB. A CI's record will include all of the relevant data, such as manufacturer, vendor, location, etc. Configuration items can be created or maintained either using tables, lists, and forms within the platform, or using the Discovery application.

Consumable Assets – Low risk hardware that is not tracked by unique identifier and is of relatively low value to the organization

Disposal Vendor – Vendor authorized by Advocate Aurora Health to dispose of assets.

Deployment and Redeployment – The process responsible for configuring, assigning, and distributing hardware assets into the production environment.

Disposal – The process responsible for tracking and documenting hardware asset destruction by engaging a disposal vendor and receiving a certificate of disposal.

Management/ Operation -The process responsible for monitoring, controlling, and repairing hardware assets in the production environment and stockroom locations.

Reclamation - Recovery of off-site Hardware Assets from a Remote Worker.

High Risk Hardware – Inclusive of: Desktops, mobile devices, network equipment, storage equipment, servers, firewalls, printers, encrypted flash drives, hard drives, and with data stored containing Information Classification 3 or 4 as defined in the AAH Information Classification Policy.

Incident - An unplanned interruption or reduction in the quality of a HIT service, system, or asset for the customer. An Incident can include:

- If a team member is unable to perform their job.
- Team member requires training to do their job and Service Desk supplies it.
- Something is broken and needs to be fixed.
- Issues where we redirect because something is broken (Clinical Engineering, Sunquest, Etc.).
- A System isn't performing as expected.

Low Risk Hardware - Inclusive of: Accessories, cables, carts, batteries & power cords, monitors, digital sign tracking boards, laptop mounts, keyboards, medical imaging graphic cards, phones, headsets, webcams, and mounts.

Non-Consumable Assets – Assets tracked by a unique identifier

Nonstandard Procurement /Acquisition - The process responsible for obtaining approvals and quotes resulting in the purchase of equipment that is not part of our standard catalog of IT items we order or stock.

Problem - Are tasks assigned to various team members to identify known errors, workarounds and to perform root cause analysis. Tasks can be considered action items.

Procurement/ Acquisition - The process responsible for purchasing hardware assets.

Reclamation – Any change in status that necessitates AAH in reclaiming the equipment.

Reorder Point -Predetermined number of units required to initiate a system inventory order to restore the required quantity of equipment on hand in a stockroom.

Reorder Quantity - The number of units AAH orders from a supplier to maintain adequate inventory in the stockrooms.

Request -Used to obtain a HIT service, system, asset or information which the customer did not have previously. A Request can include:

- Password Resets
- Adding a new printer to the workstation
- New Software Installs
- System Changes – Upgrades – Enhancements
- Equipment Moves
- Issues where we redirect for Authorization

Request Item -Line item of info from the item in the team member's cart. This will have specific properties fields filled out by the team member. This is what they want.

Request Task-Component that IT works on to do their part in fulfilling the request. There can be multiple tasks to fulfill one request and this will spawn a separate task for each team to do their part before the request can be considered closed.

Standard Hardware – Hardware configurations approved by the HIT Business Office and HIT Technical teams.

Stockroom -A secure centralized storage area for IT equipment.

Subcontractor - Subcontractor (also known as Third Party): An entity or person other than the vendor that offers to furnish goods or services of any kind under the contract.

Unique Identifier: - The serial number or other identifier assigned to the hardware by the manufacturer and/or AAH that uniquely identifies the item.

Useful Life -The timeframe that signifies the functional and financial value deterioration of a hardware asset.

Vendor - An external party providing paid goods or services to Advocate Aurora Health Care.

IV. STANDARDS

A. Procurement /Acquisition

1. Non-Consumable/High Risk Assets

- Equipment procurement/Acquisition is coordinated by the Business office HIT Equipment Buyer.
- Equipment distribution is coordinated by the Business office HIT Procurement Manager.
- Every asset must have an asset tag.
- Every asset must have a unique identifier assigned by the manufacturer.
- Every asset must have a related purchase order and request number.

- f. Every asset that may contain data identified as Information Classification 3 or 4 per the AAH Information Classification policy should be identified in the Asset Management system.
- g. Every asset that may contain data identified as Information Classification 3 or 4 per the AAH Information Classification policy in which storage is removed the storage must be tracked as a separate asset.

2. Consumable/Low Risk Assets

- a. Equipment procurement/Acquisition is coordinated by the Business office HIT Equipment Buyer.
- b. Equipment distribution is coordinated by the Business office HIT Procurement Manager.
- c. Tracked by quantity only

3. Procurement /Acquisition of Non-Standard Items

- a. Requires additional approval from the HIT Asset Management team/ HIT Field Services team members.
- b. Requires a Non-Standard Hardware Quote Request.
- c. Requires Business Owner funding approval.

B. Deployment and Redeployment

- a. AAH owned desktops and laptops must use an Advocate Aurora standard image, including:
 - i. Use Advocate Aurora Disk encryption - The disk encryption solution must validate that the desktop or laptop is encrypted and is operating correctly.
 - ii. Use Advocate Aurora antivirus software with malware signature, heuristic, anti-spyware and reputation awareness capabilities. Vendor owned or provided desktops and laptops must also use the Advocate Aurora antivirus software—if it is not technically possible, HIT Security must review the antivirus to ensure it provides adequate protection.
 - iii. Must automatically restart if the desktop or laptop is not restarted within 7 days of a patch release, unless otherwise excepted.
 - iv. Use current, up-to-date patches on operating systems. Applications shall be updated on an as-needed basis based on business case.
 - v. Include firewalls on desktops and laptops running Windows 10 or newer Windows operating systems. Desktops and laptops running other operating systems must have firewalls enabled where technically feasible.
 - vi. Use Advocate Aurora Host Intrusion Prevention System (HIPS), where technically feasible.

- vii. Automatically lockout after a maximum of 15 minutes of inactivity per AAH Integrated Desktop/Laptop policy
- viii. Local administrative rights are disabled by default where feasible.
- ix. Disable all identified unnecessary and unused services or ports.
- x. Be included in Advocate Aurora's centralized configuration management software. The audit capability must include an inventory of applications and the current patch level.
- xi. Use auditable remote desktop tools. These tools must only be used for a legitimate business or HIT assistance need.
- b. Evaluation of asset to determine if it is fit for use and has not exceed the defined useful life period
- c. Evaluation to determine if a chargeback should be performed. All lost, stolen, or damaged equipment will be charged back to the department associated with the business owner.

C. Management/ Operations

- a. Consumable assets may be stored in a designated unlocked area in locations associated with a stockroom.
- b. Non-consumable assets must be stored in a designated stockroom.
- c. Reorder levels for stockrooms will be controlled by the HIT Procurement Manager.
- d. Reorder levels will be monitored by the HIT Inventory Managers assigned to stockroom locations.
- e. All end user assets must have an identified owner and location documented in the Asset Management System.
- f. All shared assets must have an identified department level owner and location documented in Asset Management System.
- g. Assets that need special handling due to the risks associated such as Information Classification 3 or 4 related hardware must be placed in secured areas within the appropriate stockroom.
- h. HIT Procurement Manager is responsible for supervising the purchase of IT hardware required to ensure sufficient inventory quantities for stockrooms.
- i. Non-consumable assets that are managed by HIT found to be missing an asset management record must be added to the asset management system.
- j. Any asset removed or added to a stockroom must have the appropriate Request and Transfer Orders. All actions on an asset such as repair, loan, and removal of hard drives must be documented in a Request
- k. Any drive or other storage mechanism that may contain restricted information that is removed from an asset must be added to the system as a separate and distinct asset that is tracked through all Asset States.

D. Reclamation

- a. Separation of employment
 - i. Refer to the request workflow that includes returning equipment to the business upon separation of employment
- b. Remote Workers
 - i. AAH Remote Worker Procedure for Hardware Reclamation

E. Transfers and Divestitures

a. Transfers

- i. Refer to the request workflow: Transfer of device from AAH hardware asset inventory to non AAH team member, vendor, subcontractor or business associate.

b. Divestiture

- i. **Refer to the request workflow:** Refer to the request workflow: Divestiture of device (s) from AAH hardware asset(s) inventory to non AAH entity(s).

c. Reconciliation

- i. Inventory of transfers and divestitures of devices to non AAH team members, vendors, subcontractors, business associates, and entities must be reconciled by HIT asset management systems.

F. Disposal/Destruction

- a. Assets that may contain data identified as Information Classification 3 or 4 per the AAH Information Classification policy must adhere to the standard disposal procedure.
- b. All non-consumable assets must have an asset tag and unique identifier documented prior to shipping to the disposal vendor.
- c. All consumable assets must be tracked by quantity disposed.
- d. Assets must be marked with a state of "Disposed" only when a confirmation is received from the disposal vendor.
- e. A certificate of data destruction must be provided by the disposal vendor and attached to related Request in the Asset Management system before being set to state of disposed.

G. Reconciliation

- a. Reconciliation workflows are used to complete an in-depth review and analysis of asset records with the primary purpose of mitigating risk.
- b. Lists of devices and their types are used to verify being a "valid asset" or substantiate a need for evaluation.
- c. Information sources for reconciliation include vendor procurement data, network activity data, encryption status data, recycler/disposal data, and COTG inventory lists.
- d. Assumptions required for successful reconciliation: data sources will be normalized, only Advocate Aurora owned/leased devices are included, "one offs" will be dealt with as they are discovered.

- e. An independent physical inventory of all stockrooms and the equipment will be completed annually.

V. **Roles and Responsibilities**

<u>Role</u>	<u>Responsibilities</u>
Procurement Manager	• Analyze asset information • Updated asset models • Create new asset models and substitutes • Publish asset models into ServiceNow catalog • Generate quotes for non-standard assets • Purchase assets • Receive and validate asset shipments from vendors • Add assts into inventory • Create requests to transfer assets • Review and fulfill stockroom replenishment requests • Identify assets scheduled for retirement
Inventory Managers	• Receive and maintain/monitor available inventory • Establish minimum order quantity • Process stock rules (automated) • Request inventory for a stockroom • Approve inventory substitution • Approve transfer of inventory from stockroom
Inventory Specialist	• Receive and validate stockroom inventory shipments • Asset tag stockroom inventory shipments • Source assets from stockroom • Conduct regular counts of inventory • Yearly reconciliation count • Receive decommissioned assets and secure/hold for collection • Engage authorized recycler • Verify disposal with asset recycler • Attach decommission certificate to request
Field Services	• Validate asset requests • Process requests to move/add/change • Receive assets for deployment • Configure and deploy assets • Retrieve assets for decommission

VI. **CROSS REFERENCES**

AAH Hardware Asset Management
 AAH Integrated Desktop/Laptop Security
 AAH Fixed Asset Receiving Tagging and Tracking
 AAH Physical Inventory of Capital Assets
 AAH Fixed Asset Disposal and Transfer
 AAH Remote Worker Procedure for Hardware Reclamation
 AAH Information Classification

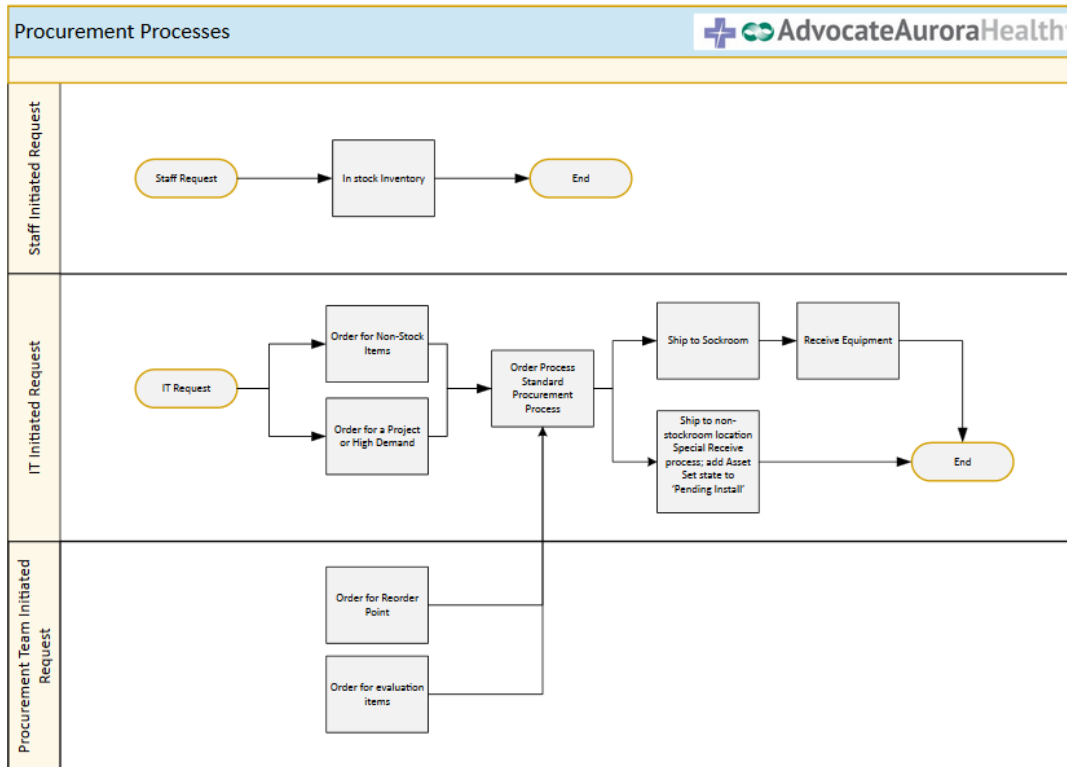
VII. RESOURCES AND REFERENCES

Lost/Stolen Workflow

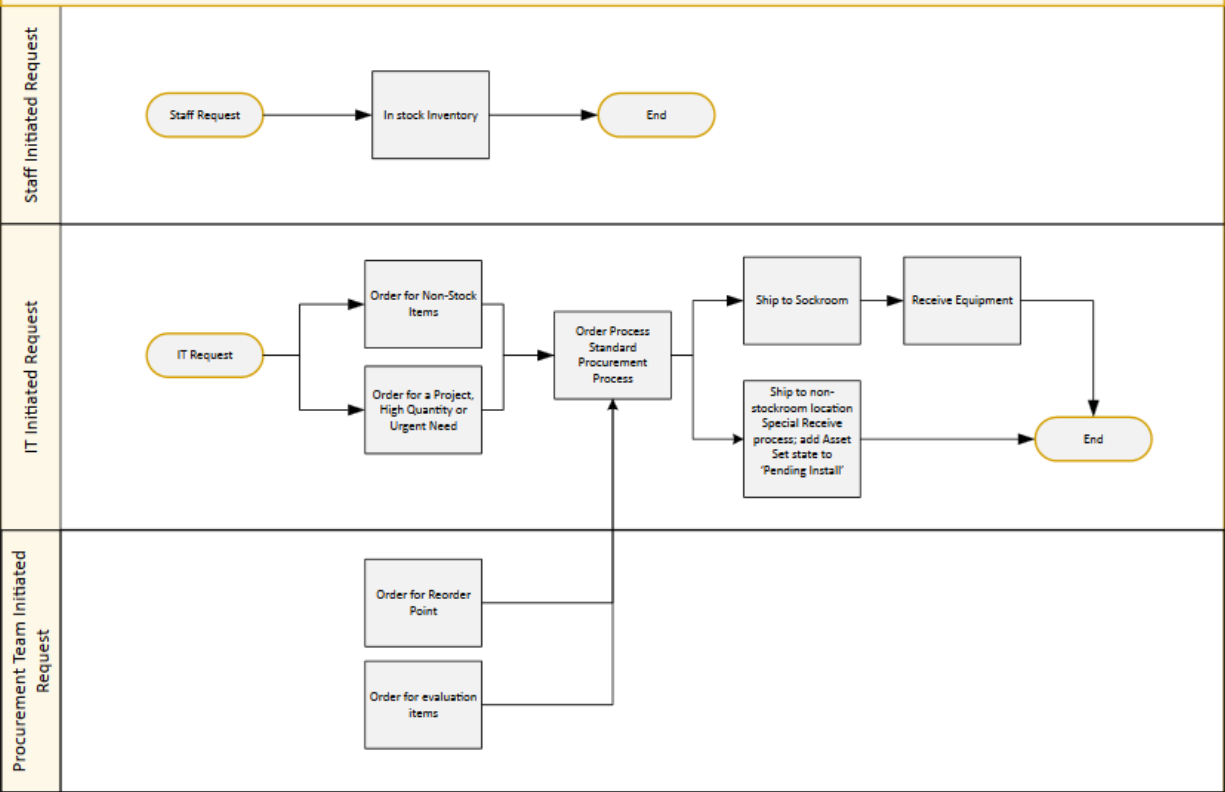
List of References:	Automation Support for Security Control Assessments v2: Hardware Asset Management; https://nvlpubs.nist.gov/nistpubs/ir/2017/NIST.IR.8011-2.pdf ; June 2017. Retrieved Dec 8, 2020. https://www.cisa.gov/sites/default/files/publications/CRR_Resource_Guide-AM.pdf Guidelines for Media Sanitization v1. https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-88r1.pdf; December 2014. Retrieved Mar 8, 2021.
Supported Compliance Controls:	
	HIPAA Security Rule: 45 cFR Part 160 and Subparts A and C of Part 164
CIS Controls v8 (May 2021)	CIS Control 1 – Inventory and Control of Enterprise Assets CIS Control 4 - Secure Configurations of Enterprise Assets and Software CIS Control 6: Access Control Management CIS Control 14: Security Awareness and Skills Training
NIST 800-53	SI-4, CM-8, IA-4

VIII. ATTACHMENTS

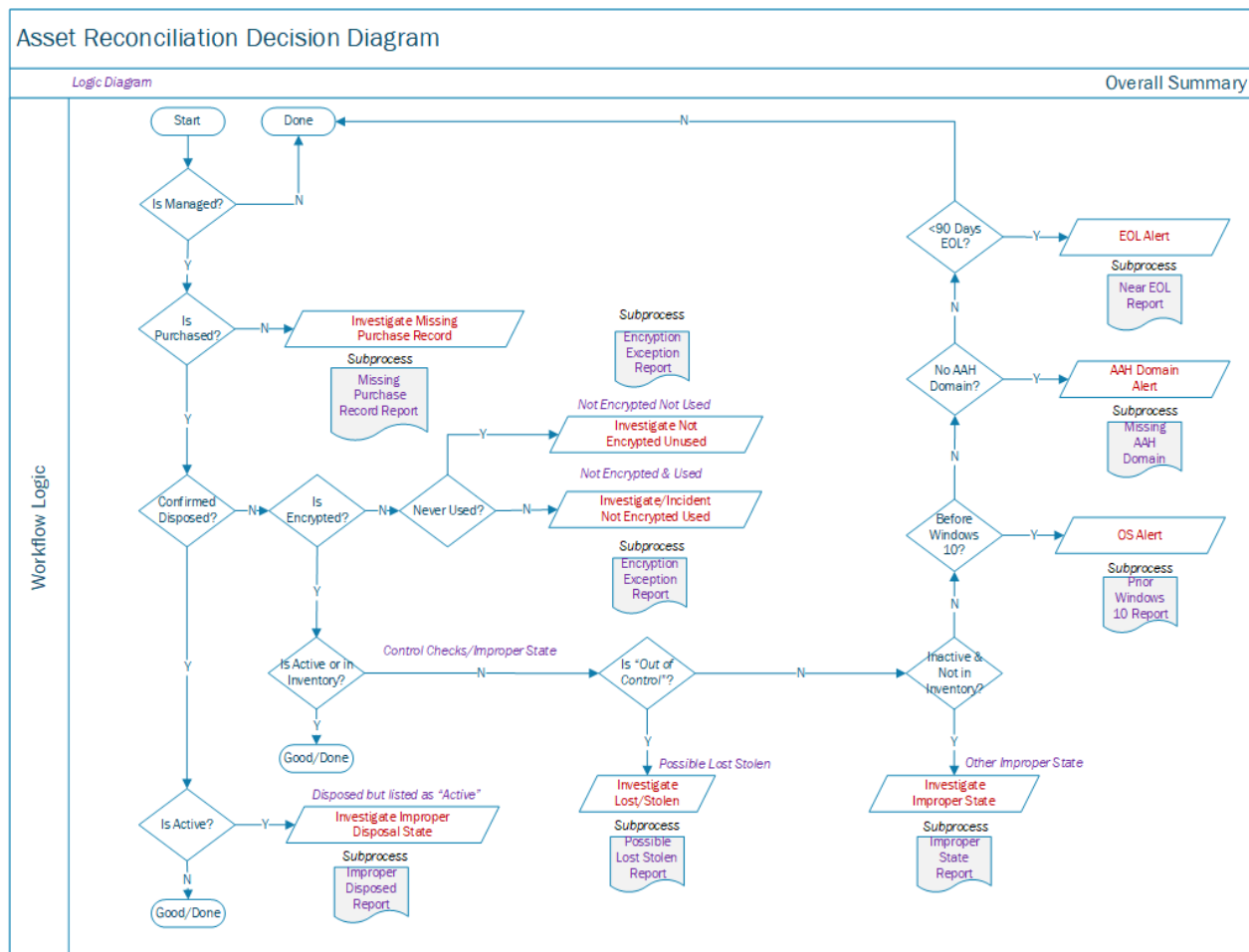
Procurement Workflow



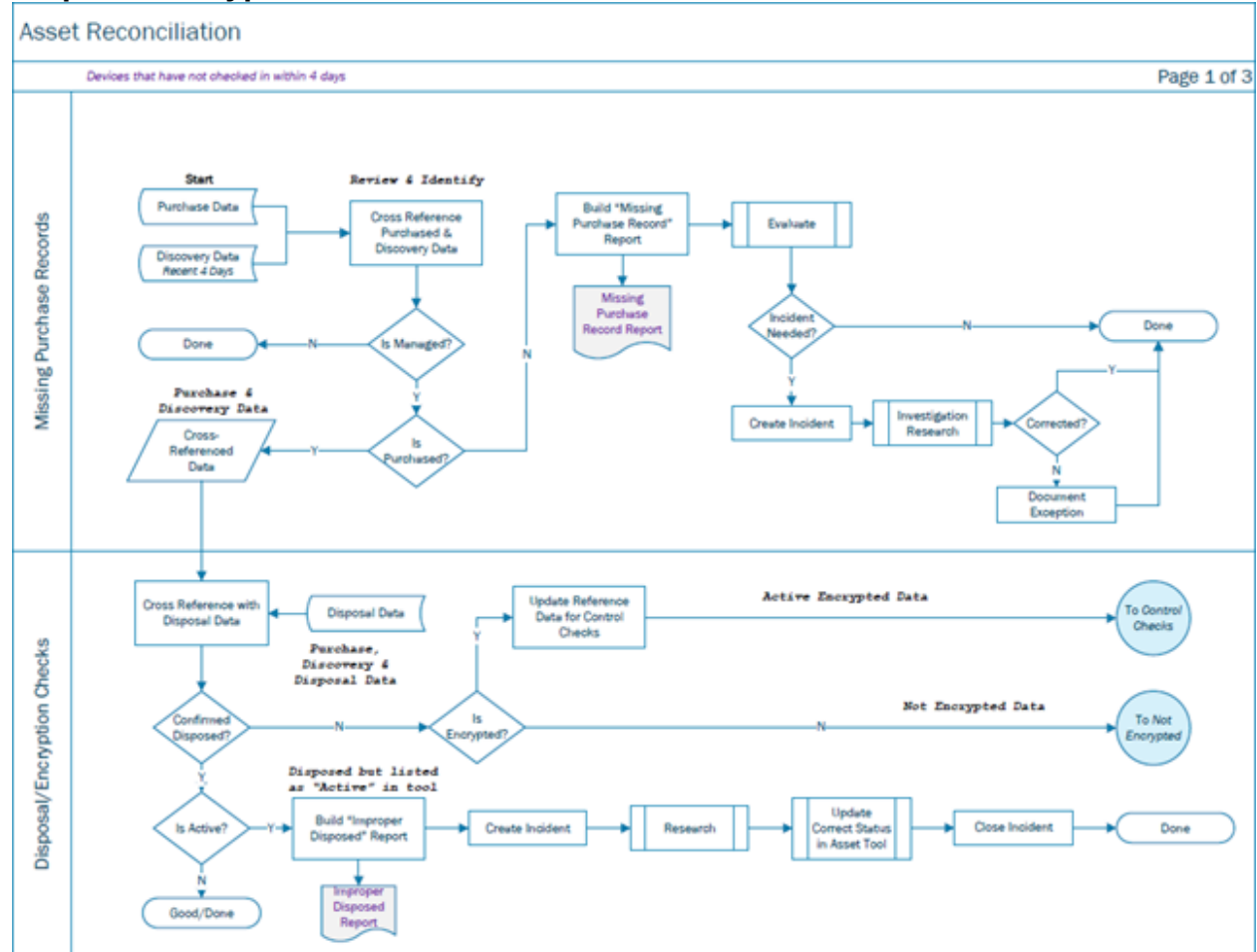
Inventory Management workflow



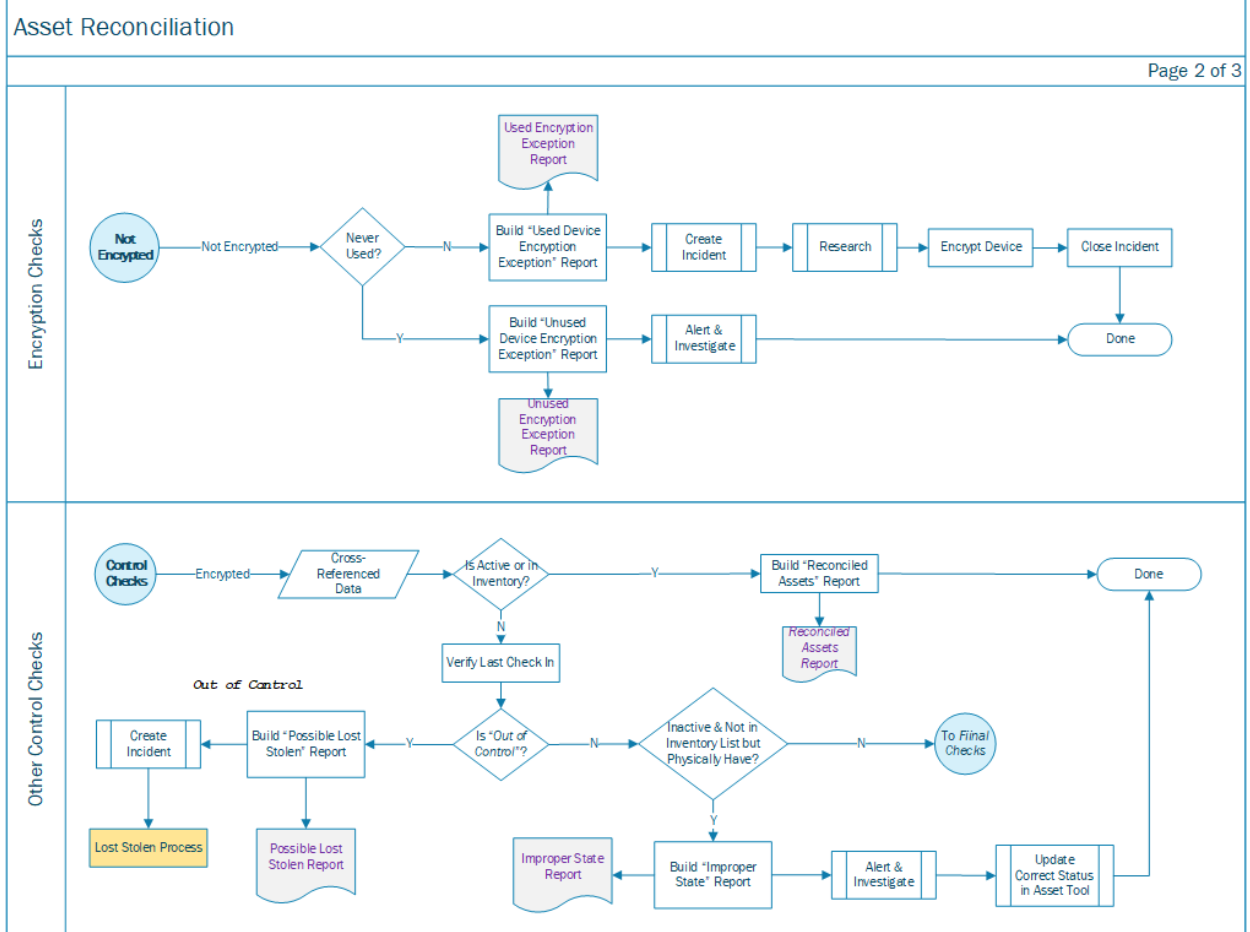
Asset Reconciliation Workflow: Logic



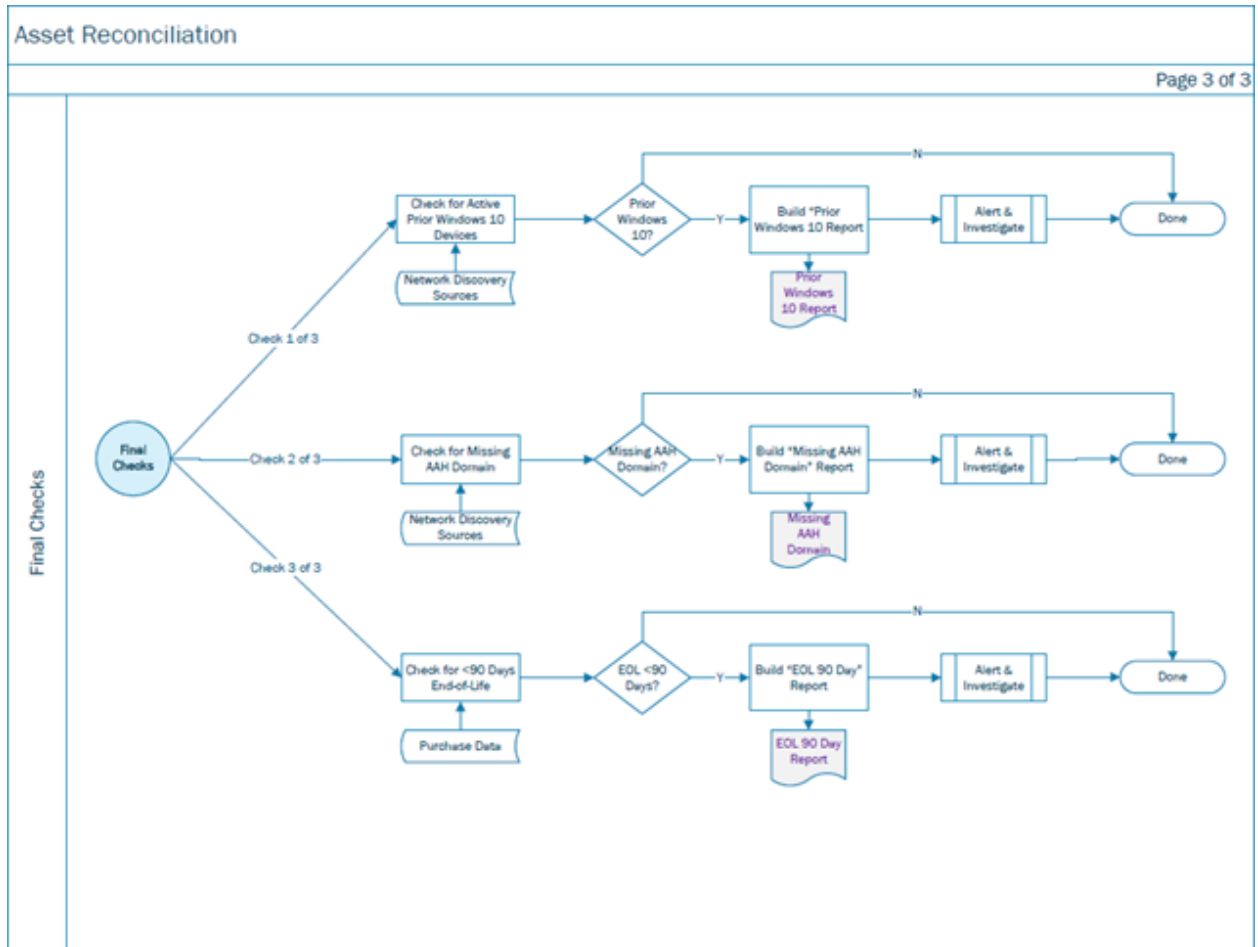
Asset Reconciliation Workflow: Missing Purchase Records and Disposal/Encryption Checks



Asset Reconciliation Workflow: Encryption and Other Control Checks



Asset Reconciliation Workflow: Final Checks



IX. APPENDICES

N/A